

1 Lab 0: Basics

1.1 Introduction

The laboratory is focused on the computation of the main parameters that affect the behavior of a high-performance vehicle. In particular, the aim is to obtain the minimum lap time on a circuit, the coefficients related to the aerodynamic forces through speed tests and the loads on the wheels while braking.

1.2 Exercise 1: Vehicle speed in a corner

Consider a car with the following characteristics:

Parameter	Value
Downforce (@ 50 m/s)	2700 N
Mass	500 kg
Friction coefficient	0.9
Radius 1	150
Radius 2	30

Table 1Exercise parameters

Using the point mass model

$$m \frac{v^2}{R} = \mu F_z \quad (1)$$

$$F_{downforce} = \frac{1}{2} \rho c_z A v^2 \quad (2)$$

$$F_z = F_{downforce} + mg \quad (3)$$

Two simulations are set up: one considering aerodynamic downforce and another without it. These configurations are evaluated on two corners with different radii: 150 m and 30 m, respectively. From the given downforce at 50 m/s, we compute the aerodynamic constant $C = \frac{1}{2} \rho c_z A$. By substituting this term into the centripetal equilibrium equation, we can solve for the maximum cornering velocity:

$$m \frac{v^2}{R} = \mu (C v^2 + mg)$$

1.2.1 Results

setup	radius	Velocity
Downforce	150 m	43.24 m/s
Downforce	30 m	16.77 m/s
NO Downforce	150 m	36.39 m/s
NO Downforce	30 m	16.27 m/s

Table 2Results

Clearly, aerodynamic downforce allows the car to reach higher cornering speeds due to the increased vertical load. However, on the tighter 30 m radius curve, the maximum speed is low enough that the downforce only contributes to a 6load. In contrast, on the wider 150 m radius curve, the higher cornering speed results in a downforce contribution of approximately 40becomes significantly more dominant as cornering speeds increase.

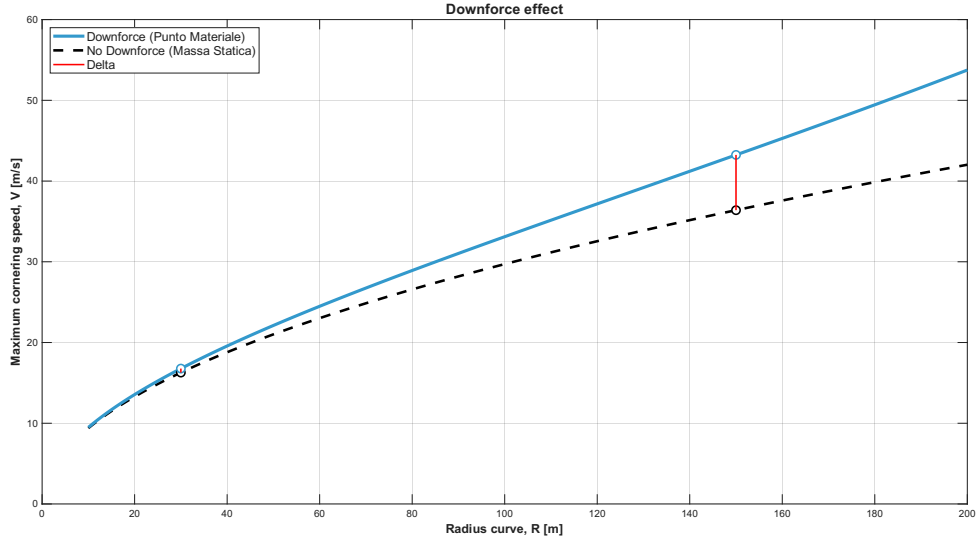


Figure 1: Dynamic effect

To further investigate this phenomenon, the calculations are repeated for a continuous range of corner radii from 10 m to 200 m. As illustrated in the figure above, the performance gap between the two setups is practically negligible at tight, low-speed corners, but widens drastically into a clear bifurcation as the curve radius increases. This visually confirms how aerodynamic grip becomes exponentially more dominant at higher speeds.

1.3 Exercise 2: Minimum lap time

The objective of this exercise is to calculate the minimum time required for a vehicle to complete a specific closed-loop track.

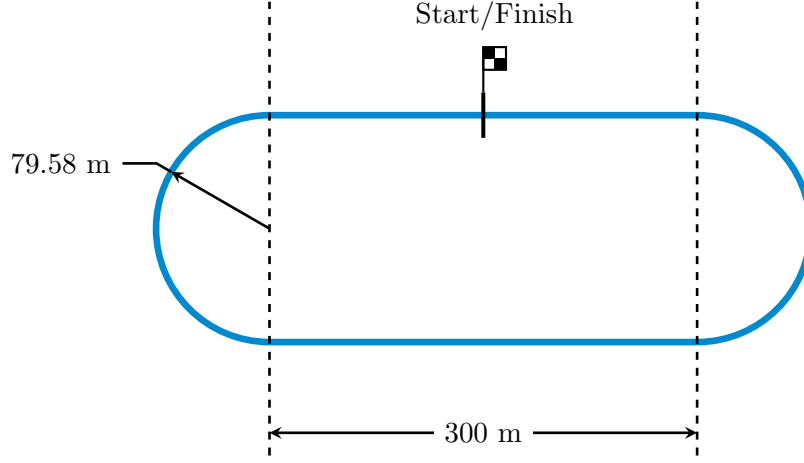


Figure 2: Track Layout

To simplify the dynamic problem, the vehicle is represented by a point mass model featuring a constant overall friction coefficient $\mu = 1.1$.

1.3.1 Optimal Driving Strategy and Speed Profile

The analysis begins by determining the maximum sustainable speed in the corners. The lateral dynamics are entirely governed by the friction limit. Therefore, the cornering speed v_{corner} is calculated by equating the required centripetal acceleration to the maximum grip:

$$m \frac{v_{corner}^2}{R_{corner}} = \mu F_z \Rightarrow v_{corner} = \sqrt{\mu R_{corner} g}$$

For the longitudinal dynamics on the straight sections, combined grip conditions are neglected. This implies that the full friction coefficient is available for purely longitudinal forces as soon as the vehicle exits the corner:

$$a_{max} = \mu g$$

The optimal driving strategy dictates that the driver must accelerate as much as possible out of the corner and brake as late as possible before the next one. Since the acceleration and braking limits are symmetrical, the peak velocity v_{max} is achieved exactly at the midpoint of the straight. The maximum speed is obtained through the standard kinematic equation for constant acceleration:

$$v_{max} = \sqrt{v_{corner}^2 + 2a_{max} \frac{L_{straight}}{2}}$$

Quantity	Value
v_{corner}	105.5 km/h
v_{max}	230.4 km/h

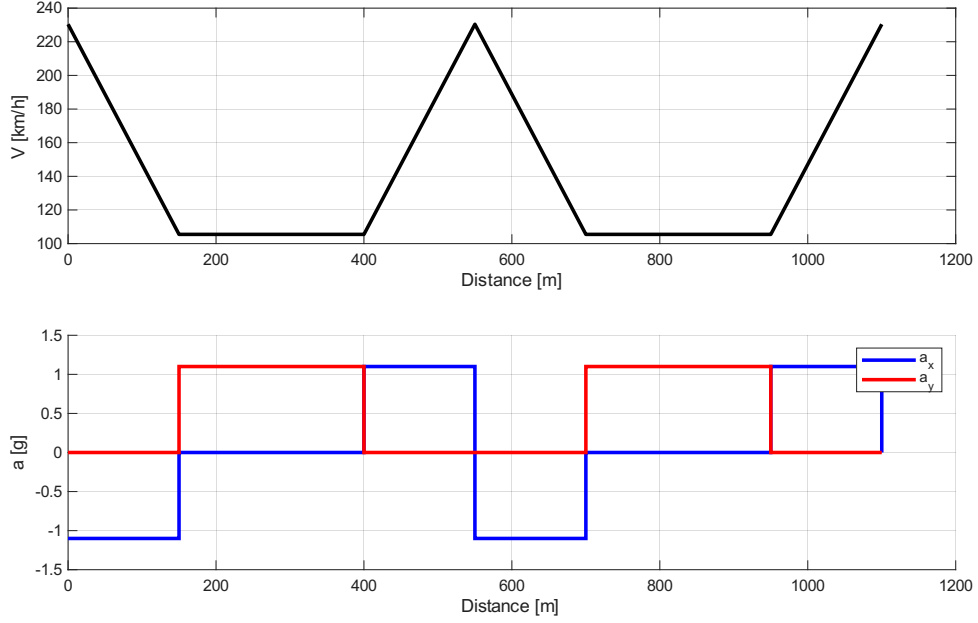


Figure 3: Lap Telemetry

By assuming a constant acceleration and deceleration bounded only by a constant friction coefficient ($\mu = 1.1$), the velocity increases linearly over time ($v(t) = v_0 + at$) up to the exact braking point, resulting in a sharp vertex.

1.3.2 Lap Time Computation: Results and Comments

To compute the total lap time, the track was divided into its sequential dynamic phases: acceleration, constant-speed cornering, and braking. The duration of each phase is calculated algebraically using the previously derived velocity thresholds and the maximum longitudinal acceleration. These individual time intervals are then arrayed according to the track layout and cumulatively summed to construct the temporal vector and determine the final lap time.

```

1 ...
2 t_straight = 2*(v_max- v_corner)/a_max;
3 t_corner = L_corner / v_corner;
4 % start/finish = middle of the straight
5 t_profile = [0, t_straight/2, t_corner, t_straight/2, t_straight/2,
6             t_corner, t_straight/2];
7 for ii = 2: length(t_profile)
8     a = t(end) + t_profile(ii);
9     t = [t a];
10 end
11 LapTime = t(end);
12 ...

```

Quantity	Value
Minimum Lap-time	29.924 s

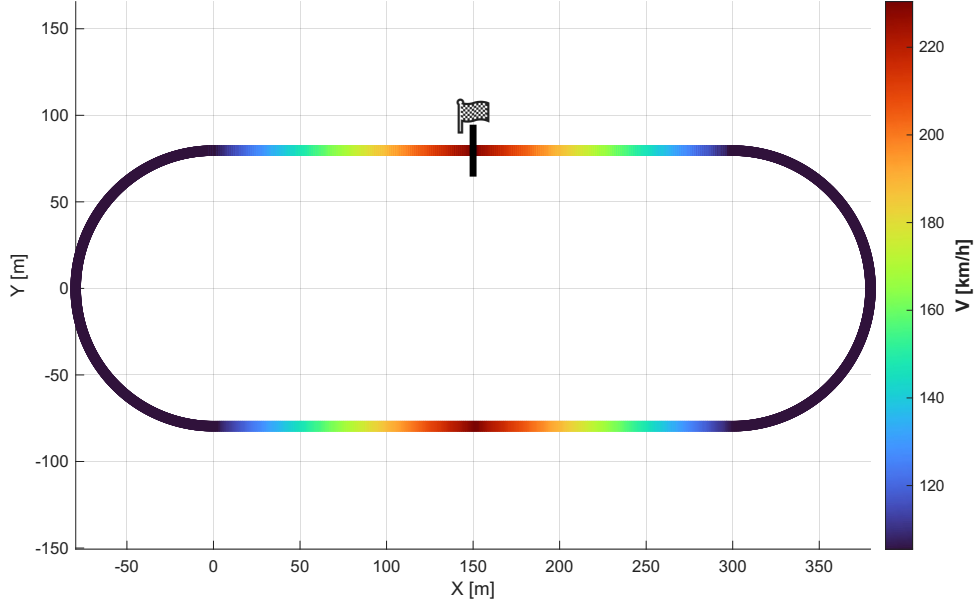


Figure 4: Speed Profile

While the point-mass kinematic approach provides a useful baseline for the minimum lap time, it relies on several physical simplifications compared to real-world racing conditions:

- **Combined Slip:** Assumes instant switching between pure longitudinal and lateral forces, ignoring the friction ellipse and the need for smooth transitions.
- **Engine Limits and Resistant Forces:** The sharp, "triangular" velocity peaks assume a constant longitudinal acceleration. In a real vehicle, acceleration is constrained by the engine power limits (F_{engine}) and tapers off at higher speeds due to the quadratic increase of aerodynamic drag ($F_{Drag} = \frac{1}{2}\rho c_x A v^2$) and rolling resistance.
- **Aerodynamic Downforce:** Neglects the generation of downforce ($F_{Downforce} = \frac{1}{2}\rho c_z A v^2$), which significantly increases vertical load and cornering limits at high speeds.
- **Internal Dynamics and Load Transfers:** Ignores the center of gravity height ($h_{c.o.g.}$) and suspension dynamics. Dynamic weight shifts naturally reduce the vehicle's total available grip due to tire load sensitivity.

1.4 Exercise 3: Coast down test

The objective of this exercise is to estimate the aerodynamic drag area ($C_x \cdot S$) and the rolling resistance coefficient (f_{roll}) of a vehicle by analyzing the velocity profile acquired during a **coast down test**. The test is a cost-effective alternative to wind tunnel testing.

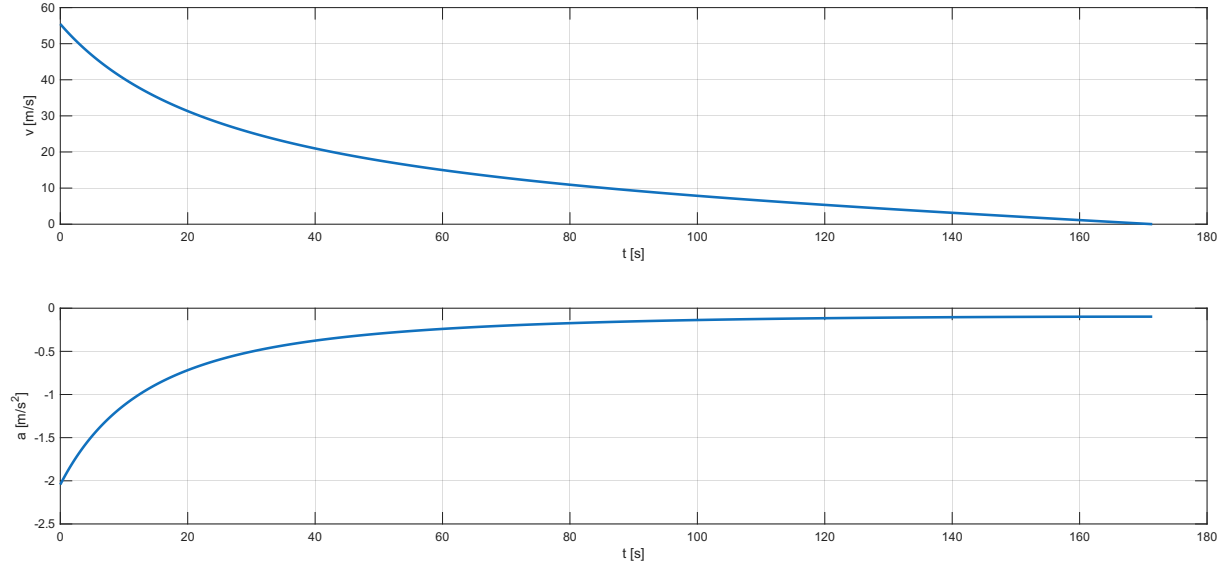


Figure 5: Speed and Acceleration Profiles

The analysis is based on a point mass model and the longitudinal dynamics equation. During a coast down test on a flat surface, both the driving force from the engine and the grade resistance are assumed to be zero ($F_{engine} = 0$, $F_{grade} = 0$). Therefore, the fundamental equation reduces to:

$$ma_x = -F_{Drag} - F_{roll}$$

Quantity	Value
Mass (m)	1380 kg
$C_z \cdot S$	2 m²

Table 3: Vehicle Data

1.4.1 Governing Equations and Curve Fitting

The individual resistive forces are defined as:

- **Aerodynamic Drag:** $F_{Drag} = \frac{1}{2}\rho c_x S v^2$
- **Rolling Resistance:** $F_{roll} = f_{roll} F_z$: the total vertical load F_z is not constant but it is the sum of the static weight of the car (mg) and the aerodynamic downforce ($F_{Downforce} = \frac{1}{2}\rho c_z S v^2$).

The total resistive force F_{res} can be expressed as:

$$F_{res} = -ma_x = f_{roll} \left(mg + \frac{1}{2}\rho c_z S v^2 \right) + \frac{1}{2}\rho c_x S v^2$$

By grouping the terms based on the square of the velocity (v^2), the equation takes the form of a second-order polynomial with respect to v :

$$F_{res} = \left[\frac{1}{2} \rho (c_x S + f_{roll} c_z S) \right] v^2 + [f_{roll} m g]$$

According to the physical model, the longitudinal resistive forces consist purely of a constant component (static rolling resistance) and a quadratic component (aerodynamic drag and downforce-dependent rolling resistance). To mathematically extract the unknown parameters (f_{roll} and $c_x S$) from the discrete dataset, an **analytical curve fitting** is required.

The regression was performed as a first-degree linear fit with respect to the square of the velocity (v^2) using MATLAB's `polyfit` function. This was implemented by squaring the velocity array and executing a linear regression, which yields exactly the equation $F_{res} = p_1 v^2 + p_2$, perfectly matching the theoretical formulation.

```

1 ...
2 v_quad = v.^2;
3 p = polyfit(v_quad, F_res, 1); % p = [p(1), p(2)]
4 % F_res = p(1) v^2 + p(2)
5 ...

```

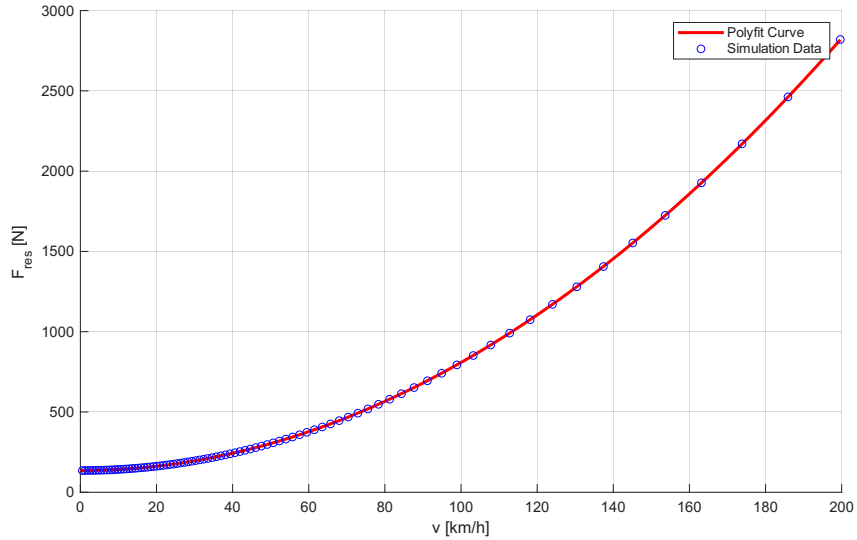


Figure 6: Comparison between simulated data and theoretical curve using `polyfit`

As shown, the theoretical model perfectly matches the simulated data points. The exact overlap confirms the validity of the assumption that the longitudinal resistance is strictly a function of the static rolling resistance and the square of the velocity.

1.4.2 Parameters Computation: Results and Comments

The desired physical coefficients are then directly derived from the polynomial outputs.

```

1 ...
2 froll = p(2)/(m*g); % [-]
3 CxS = 2*p(1)/rho_air - froll*czs; % [m^2]
4 ...

```

Quantity	Value
f_{roll}	0.01
$C_x \cdot S$	1.4044 m ²

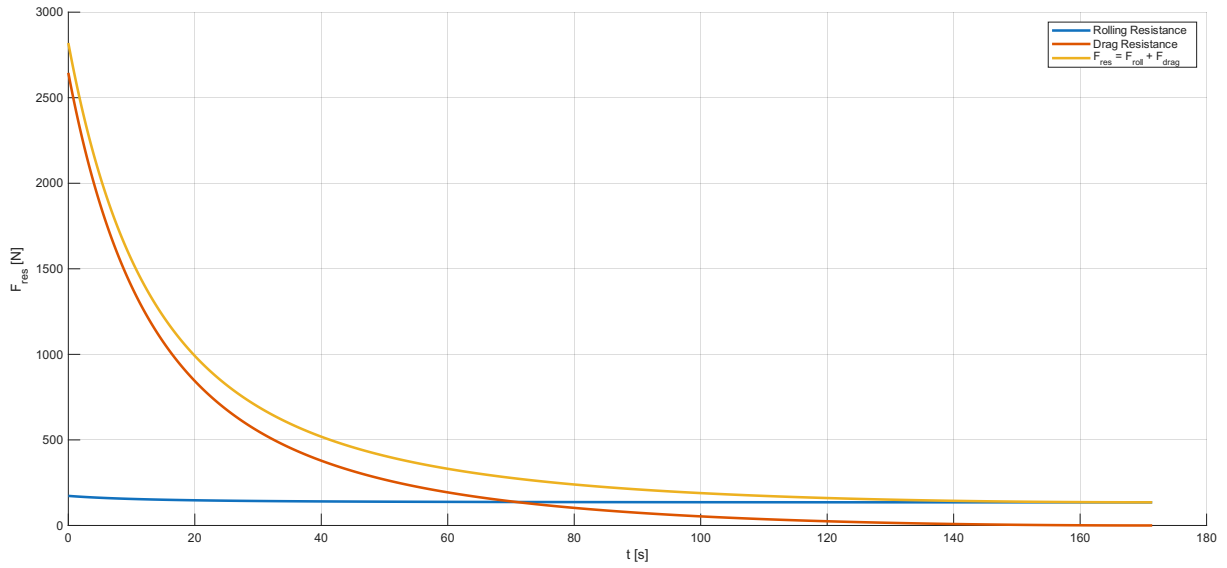


Figure 7: Resistance Forces Profile

The time-history plot highlights the relative impact of the different forces. At the beginning of the coast down (high speeds), aerodynamic drag is the dominant resistive force. As the vehicle decelerates, F_{Drag} drops quadratically. Furthermore, the plot shows that the rolling resistance F_{roll} is not perfectly constant over time; it slightly decreases as the speed drops. This is consistent with the physical model: at lower speeds, the vehicle loses aerodynamic downforce, which in turn reduces the vertical load F_z on the tires and the resulting rolling friction.

1.5 Exercise 4: Load transfer during braking

Consider a car with the following characteristics:

Quantity	Value	Quantity	Value
Mass (m)	1250 kg	Weight distribution	46% (front)
Height of CG (h)	0.35 m	Track (t)	1.6 m
$c_z \cdot S$	3.2 m ²	Aerodynamic balance	51% (front)
Wheelbase (L)	2.65 m		

Table 4: Vehicle parameters for longitudinal dynamics analysis

The vehicle speed is 200 km/h, and the longitudinal acceleration is -17 m/s^2 . The objective is to determine the vertical forces acting on each single wheel. We assume standard air density $\rho = 1.225 \text{ kg/m}^3$ and gravity $g = 9.81 \text{ m/s}^2$. The vehicle speed is converted to $v = 55.56 \text{ m/s}$.

The total weight of the vehicle is $W = m \cdot g = 12262.5 \text{ N}$. Using the static weight distribution (46% front), the static loads per axle are:

$$W_f = W \cdot 0.46 = 5640.75 \text{ N} \quad (4)$$

$$W_r = W \cdot (1 - 0.46) = 6621.75 \text{ N} \quad (5)$$

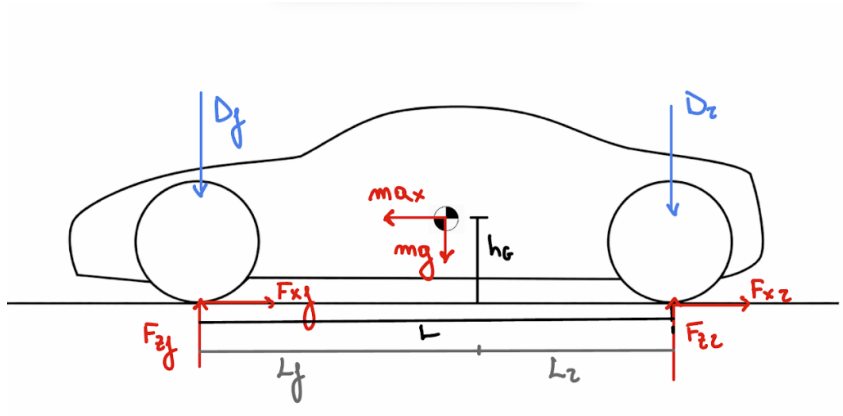


Figure 8: Dynamic effect

The total aerodynamic downforce is computed and split according to the aerodynamic balance (51% front):

$$F_D = \frac{1}{2} \rho (c_z \cdot S) v^2 = 6049.38 \text{ N} \quad (6)$$

$$D_f = F_D \cdot 0.51 = 3085.18 \text{ N} \quad (7)$$

$$D_r = F_D \cdot (1 - 0.51) = 2964.20 \text{ N} \quad (8)$$

Due to the braking deceleration, load transfers from the rear to the front axle. The transferred load ΔF_z is:

$$\Delta F_z = m \cdot |a_x| \cdot \frac{h}{L} = 1250 \cdot 17 \cdot \frac{0.35}{2.65} = 2806.60 \text{ N} \quad (9)$$

Assuming lateral symmetry (no lateral acceleration), the total vertical force on each single wheel is half the sum of the components on its respective axle.

For a single **front wheel**:

$$F_{z,\text{front}} = \frac{W_f + D_f + \Delta F_z}{2} = 5766.27 \text{ N} \quad (10)$$

For a single **rear wheel**:

$$F_{z,\text{rear}} = \frac{W_r + D_r - \Delta F_z}{2} = 3389.68 \text{ N} \quad (11)$$

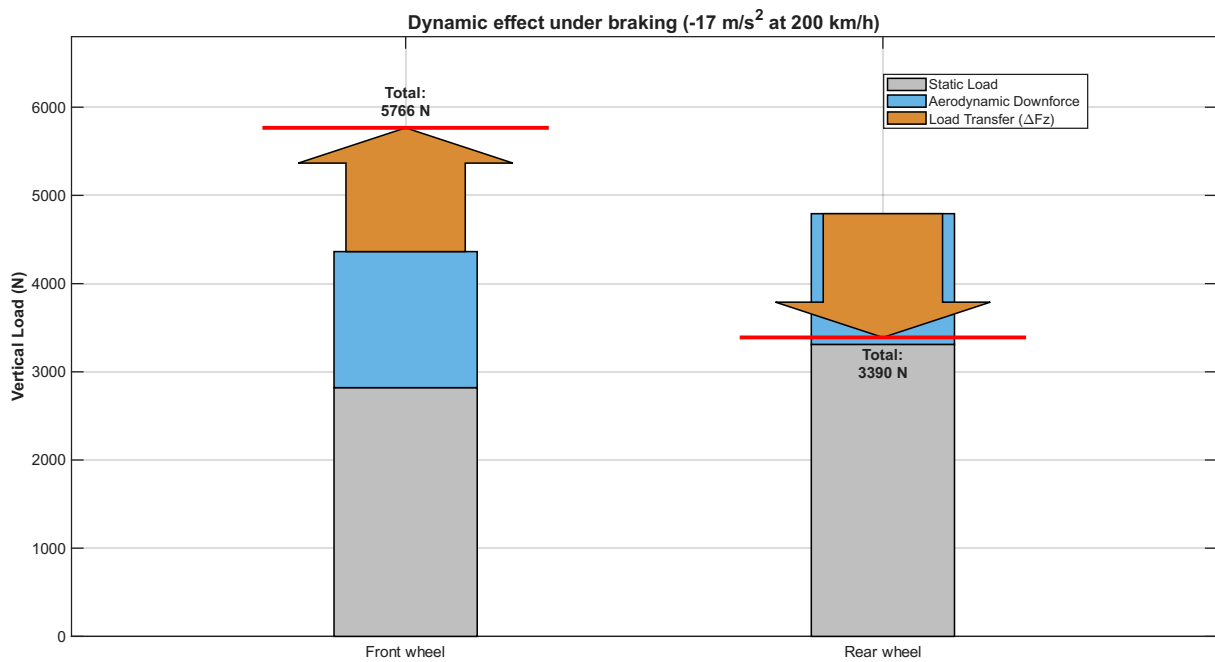


Figure 9: Dynamic effect

1.6 Exercise 5: Constant speed test

1.6.1 Methodology and main features

The **Constant Speed Test** is a practical and cost-effective on-track method used to evaluate a vehicle's aerodynamic performance, specifically to estimate the **downforce coefficient** (C_z).

The test is conducted by maintaining the vehicle at a constant speed to isolate aerodynamic loads from inertial effects. By using **suspension sensors**, it is possible to measure the **vertical load changes** at specific speeds.

An example of the data provided by the test is reported in the picture below.

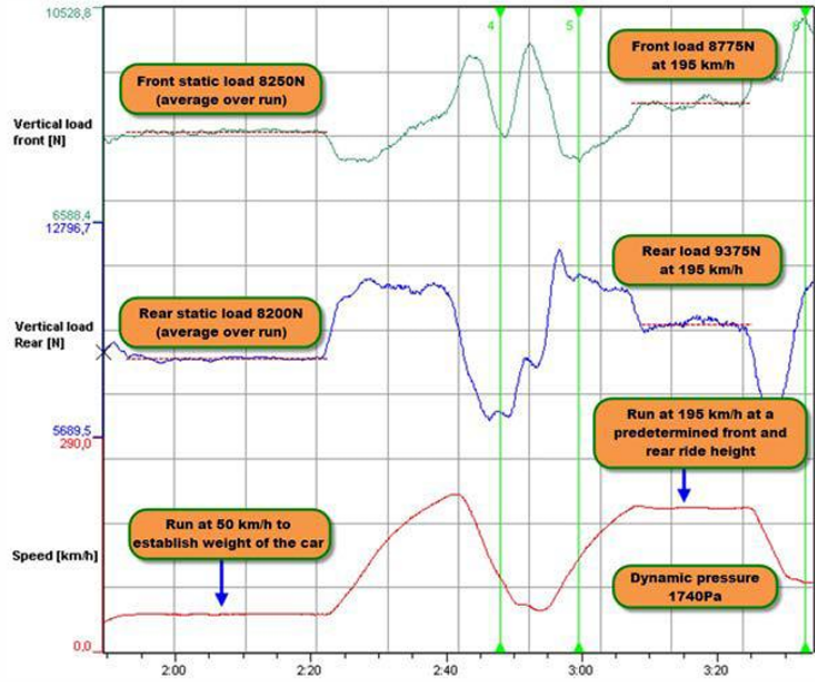


Figure 10: Constant speed test plot

As shown, the test consists of a **low-speed run phase** used to determine the **static baseline weight and ride height**. Then, the vehicle's speed is increased to perform a **high-velocity run phase** at a predetermined ride height to measure the **increased vertical load** on both the front and rear axles.

This method serves as a **cheap alternative** to wind tunnel testing. Nevertheless, the test shares many limitations with "coast down" testing, such as sensitivity to ambient wind conditions, track surface irregularities, and the difficulty of maintaining perfectly constant environmental variables.

1.6.2 Computation procedure

The aim of the test is to calculate the **aerodynamic coefficients** $C_x S$ and $C_y S$, the **aerodynamic balance** Bal and the **efficiency** Eff .

The computation is performed according to following data related to the vehicle setup and the environment.

Parameter	Value
Front spring rate $K_{s,f}$	350 N/mm
Rear spring rate $K_{s,r}$	200 N/mm
Front motion ratio $R_{m,f}$	1.82
Rear motion ratio $R_{m,r}$	1.3
Tyre rolling radius R_{rol}	345 mm
Total gear ratio τ_{tot}	4.45
Current engine torque $T(\omega)$	360 Nm
Vehicle mass m	1380 kg
Air density ρ	1.225 kg/m ³
Rolling resistance coefficient f_{roll}	0.011
Vehicle speed v	250 km/h
Front suspension travel Δz_f	6.4 mm
Rear suspension travel Δz_r	11.5 mm

Table 5: Parameters of vehicle setup

It is important to note that the front and rear suspension travel are zeroed in static position. This means that the values reported in the table refers only to the downforce effect on the vehicle.

So it is possible to obtain the **downforce** $F_{downforce}$ through the relationship between the **force on the suspension spring** F_s and the **load on the wheel** F_w , considering that:

$$F_{downforce} = 2F_{w,f} + 2F_{w,r} \quad (12)$$

where $F_{w,f}$ is the **force on a single front wheel** and $F_{w,r}$ is the **force on a single rear wheel**.

The load on the wheels is calculated by evaluating the **dynamics of the suspension structure**, represented in the following figure. reported in the picture below.

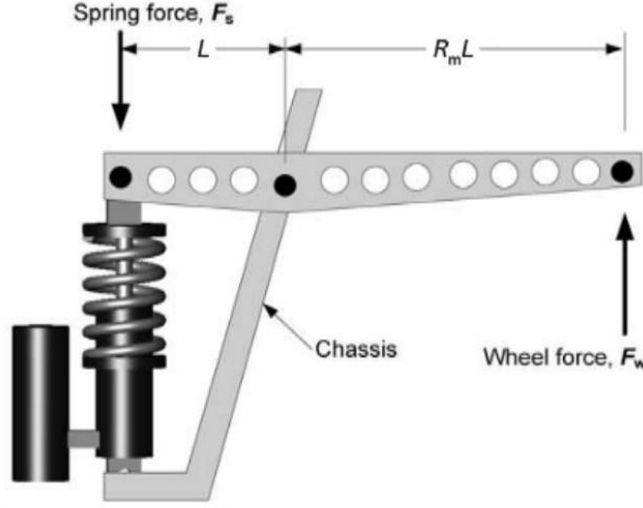


Figure 11: Suspension structure

Applying the equilibrium of the moments given by the forces, the final relationship between F_w and F_s is reported in the formula below.

$$F_w = \frac{F_s}{R_m} = \frac{K_s \Delta z}{R_m} \quad (13)$$

Then, the **downforce area** $C_z S$ is obtained according to the following equation.

$$C_z S = \frac{2F_{down,force}}{\rho v^2} \quad (14)$$

After that, it is possible to compute the **aerodynamic balance** Bal defined as:

$$Bal = \frac{F_{down,force,front}}{F_{down,force}} = \frac{2F_{w,f}}{F_{down,force}} \quad (15)$$

The **drag force** F_{drag} is computed by analyzing the **longitudinal dynamics** of the vehicle.

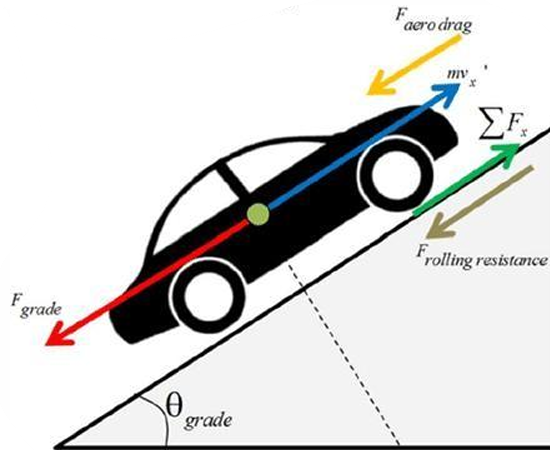


Figure 12: Longitudinal dynamics of a vehicle

As shown in the picture above, the **drag force** F_{drag} is given by the equilibrium of the longitudinal forces.

$$F_{drag} = F_{engine} - ma_x - F_{roll} - F_{grade} \quad (16)$$

where F_{engine} is the **traction force**, a_x is the **longitudinal acceleration** expressed in m/s^2 , F_{roll} is the **rolling resistance** and F_{grade} is the **resistance force related to the slope of the road**.

Since the vehicle runs at constant speed on level ground, a_x and F_{grade} are null.

The remaining forces are computed via the following equations.

$$F_{engine} = \frac{\tau_{tot}T(\omega)}{R_{rol}} \quad (17)$$

$$F_{roll} = f_{roll}F_z = f_{roll}(mg + F_{downforce}) \quad (18)$$

Finally, the **drag area** C_xS and the **aerodynamic efficiency** Eff are obtained in this way:

$$C_xS = \frac{2F_{drag}}{\rho v^2} \quad (19)$$

$$Eff = \frac{C_zS}{C_xS} \quad (20)$$

The final results are reported in the following table.

Parameter	Value
downforce area C_zS	2.03 m ²
drag area C_xS	1.5 m ²
aerodynamic balance Bal	41%
aerodynamic efficiency Eff	1.35

Table 6: Final results